

Om Gawande

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Summary

Artificial Intelligence undergraduate with practical experience in building real-world computer vision and NLP systems and deploying them on embedded and fully offline infrastructure. Experienced in training custom object detection models, developing local LLM-based agentic pipelines, and delivering end-to-end AI solutions from dataset preparation to production deployment. Interested in applied AI, computer vision, and scalable intelligent systems.

Experience

Data Analytics and AI Engineering Intern — Acube AI

- Developed AI-powered applications by integrating machine learning models, LLMs, and backend APIs into production-ready systems.
- Designed prompt engineering workflows, automated business processes, and collaborated on end-to-end AI solution deployment.

AI and ERP Solution Intern — RNR

- Developed AI modules for enterprise ERP applications, including machine learning pipelines, NLP workflows, and intelligent business automation features.
- Worked with cross-functional teams to design, implement, test, and deploy scalable AI solutions for real-world business use cases.

Skills

Programming Languages: Python, SQL

Deep Learning: CNNs, YOLO object detection, transfer learning, custom model training, TensorFlow, Keras

Computer Vision: OpenCV, image preprocessing, dataset annotation, real-time detection, camera integration

Generative AI: LLM applications, RAG pipelines, Ollama, vector databases (Qdrant), prompt engineering, agentic workflow design

Libraries and Tools: NumPy, Pandas, Matplotlib, Scikit-learn, Docker, Google Colab, Git, GitHub

Embedded AI and Hardware: Raspberry Pi 4, camera module integration, IoT

Soft Skills: Leadership, Problem Solving, Strategic Thinking, Teamwork, Adaptability

Projects

Local Area Network Agentic RAG System for Institutional Knowledge Base

- Built a LAN-based retrieval-augmented generation system using Ollama-hosted LLMs, Qdrant vector search, and MySQL, enabling secure on-premise deployment without external cloud services.
- Designed a multi-agent architecture (Router, SQL, RAG, Synthesis) with a dedicated verification agent for hallucination detection and answer grounding.
- Deployed via Docker Compose with a Django/DRF backend and React frontend for end-to-end query handling across the local network.

Smart AI Waste Segregation System using YOLO and Raspberry Pi

- Built an embedded YOLOv8-based computer vision system on Raspberry Pi for real-time autonomous waste classification and segregation.
- Handled dataset annotation, custom model training, and optimization for constrained edge hardware.

Sarcasm Detection for Low-Resource Languages

- Developing a transformer-based NLP framework for sarcasm detection in regional Indian languages using custom annotated datasets.
- Defined annotation guidelines, dataset schema, and strict separation of authentic and synthetic data for research integrity.

Education

B.Tech in Artificial Intelligence (2027)

St. Vincent Pallotti College of Engineering and Technology, Nagpur

CGPA: 8.68 / 10

Certifications & Achievements

- President, AI Prabodha Student Forum (previously Secretary & Founding Member)
- Class Representative (2023–Present)
- Fundamentals of Deep Learning — NVIDIA
- Introduction to Machine Learning — NPTEL
- Data Engineering in Cloud Environment Training
- Generative AI & DevOps Faculty Development Program